

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

# ICT Level 1

## Teaching Guide

Level: Level 1

Class: Class 6

Duration: June - January (8 months)

Period: 40 minutes

ICT Teaching Resources Portal

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## Curriculum Overview

ICT Level 1 is designed for Class 6 students in KREIS residential schools. The curriculum follows the 5E instructional model and integrates Singapore's Concrete-Pictorial-Abstract approach for effective learning.

## Themes Covered

| Chapter | Theme                            | Periods | Key Topics                                                             |
|---------|----------------------------------|---------|------------------------------------------------------------------------|
| 1       | Internet and ICT Environment     | 12      | Introduction to ICT, Internet basics,                                  |
| 2       | Graphics in ICT                  | 12      | Web browsers, GPS<br>Painting/Drawing, Graphics                        |
| 3       | Audio & Visual Communication     | 12      | software, Symmetric designs<br>Communication types, Audio/             |
| 4       | Data Representation & Processing | 12      | Video conferencing, Audacity<br>Number systems, Data                   |
| 5       | Programming                      | 12      | representation, Basic processing<br>Scratch programming, Basic blocks, |
| 6       | Software Applications            | 12      | Simple animations,<br>Introduction to software, Types of<br>software   |

## Pedagogical Approach

### 5E Model Implementation

| Phase     | Duration  | Activities                                                                                          |
|-----------|-----------|-----------------------------------------------------------------------------------------------------|
| Engage    | 5-7 min   | Real-world connection, regional metaphors,                                                          |
| Explore   | 10-12 min | questions                                                                                           |
| Explain   | 10-12 min | Hands-on activity, Driver-Navigator pairs, Lab work                                                 |
| Elaborate | 8-10 min  | Live coding, Concept introduction, C-P-A approach                                                   |
| Evaluate  | 5-8 min   | Extended activity, Parson's problems, Application<br>Formative check, Exit tickets, Self-assessment |

### International Pedagogies Used

- **Singapore C-P-A:** Concrete manipulatives → Visual blocks → Abstract code
- **UK NCCE:** Unplugged Computing, Live Coding, Parson's Problems
- **Driver-Navigator:** Pair programming for lab sessions
- **Rural Scaffolding:** Regional metaphors (grain sorting, bus routes, hostel inventories)

### Lesson Plan Structure (40 Minutes)

| Time      | Phase     | Teacher Activity                                                     | Student Activity                                                 |
|-----------|-----------|----------------------------------------------------------------------|------------------------------------------------------------------|
| 0-5 min   | Engage    | Present real-world scenario using                                    | Discuss, share observations                                      |
| 5-15 min  | Explore   | regional metaphor<br>Guide hands-on exploration in                   | Work in Driver-Navigator pairs,                                  |
| 15-25 min | Explain   | pairs<br>Live coding demonstration,                                  | explore software<br>Follow along, ask questions, take            |
| 25-35 min | Elaborate | introduce concepts<br>Assign extension activity or                   | notes<br>Apply learning, solve problems                          |
| 35-40 min | Evaluate  | Parson's problem<br>Conduct formative assessment,<br>assign homework | independently<br>Complete exit ticket, receive<br>dormitory task |

## Assessment Framework

### Assessment Components (Per Class)

| Assessment Type          | Count     | Marks Each | Total      |
|--------------------------|-----------|------------|------------|
| Formative (FA-1 to FA-4) | 4         | 30         | 120        |
| Unit Tests               | 4         | 10         | 40         |
| Summative (SA-1, SA-2)   | 2         | 40         | 80         |
| Practical Lab Exams      | 2         | 20         | 40         |
| <b>Grand Total</b>       | <b>12</b> |            | <b>280</b> |

## Resources Required

### Hardware

- Computer lab with 1 PC per 2 students (minimum)
- Internet connection (2 Mbps minimum)
- Projector for demonstrations
- Speakers and headphones

### Software (Free/Open Source)

- Operating System: Ubuntu Linux or Windows
- Office Suite: LibreOffice / Google Workspace
- Graphics: GIMP, Tux Paint
- Audio: Audacity
- Programming: Scratch
- Browser: Firefox / Chromium

## Dormitory Tasks (Zero Lab Access)

For residential schools with limited lab time, these paper-and-pencil tasks can be assigned for supervised evening study:

- **Code Dry-Running:** Trace Scratch programs on paper
- **Variable State Tables:** Track variable values through program execution
- **Flowchart Drawing:** Create flowcharts for algorithms
- **Binary Conversion:** Practice decimal to binary conversions
- **Keyboard Layout:** Practice typing without computer
- **Software Matching:** Match software to their purposes

## Capstone Projects

### Project 1: School Environment Monitor

**Integration:** Science + ICT

**Description:** Students create a Scratch program that simulates monitoring temperature, humidity, and weather conditions in the school campus.

**Skills:** Data representation, variables, loops, conditionals

### Project 2: Hostel Store Inventory

**Integration:** Mathematics + ICT

**Description:** Students create a spreadsheet to track hostel store items (rice, dal, sugar) with quantity, rate, and total calculations.

**Skills:** Spreadsheet formulas, data entry, basic calculations

## Unplugged Activities Library

These activities teach computational thinking without any computer access. Use them in regular classrooms, during lab shortages, or as warm-up exercises.

### Activity 1: Human Robot (Ch 5 - Programming)

**Duration:** 15 minutes | **Materials:** None

#### Steps:

1. One student is the “robot”, another is the “programmer”
2. The programmer gives step-by-step commands: “Move forward 3 steps”, “Turn left”, “Pick up the pencil”
3. The robot follows ONLY exact instructions (cannot interpret vague commands)
4. Switch roles. Discuss: What happens when instructions are unclear?

**CT Skill:** Algorithm design, precision in communication

### **Activity 2: Packet Relay (Ch 1 - Internet)**

**Duration:** 20 minutes | **Materials:** Paper slips, tape

**Steps:**

1. Students stand in a line (the “network”)
2. Each student is a “router” who receives a message (paper slip) and passes it forward
3. If a student receives a damaged message (crumpled slip), they must request retransmission
4. Time how long the message takes to travel. Discuss: How does the internet work?

**CT Skill:** Decomposition, understanding networks

### **Activity 3: Pixel Art Grid (Ch 2 - Graphics)**

**Duration:** 15 minutes | **Materials:** Graph paper, colored pencils

**Steps:**

1. Give students a small grid (8x8) and ask them to fill squares to create a simple image
2. Use only 4 colors. Discuss: How is this like a digital image?
3. Count total pixels used. What happens if you use a bigger grid (16x16)?

**CT Skill:** Abstraction, pattern recognition

### **Activity 4: Sorting Race (Ch 4 - Data)**

**Duration:** 15 minutes | **Materials:** 30 mixed beans/grains, paper cups

**Steps:**

1. Give each pair a cup of mixed grains (rice, dal, mustard seeds)
2. Race to sort them into separate groups
3. Try different sorting strategies: one-by-one vs. grouping by type first
4. Discuss: Which strategy was faster? Why?

**CT Skill:** Pattern recognition, algorithm efficiency

### **Activity 5: Debugging Challenge (Ch 5 - Programming)**

**Duration:** 10 minutes | **Materials:** Printed “code” on paper

**Steps:**

1. Give students a simple Scratch-like program with 3 intentional errors
2. Students must find and fix each error
3. First to find all errors wins
4. Discuss: How do you find mistakes in your own work?

**CT Skill:** Debugging, logical reasoning

### **Activity 6: Story Sequencing (Ch 3 - Audio/Visual)**

**Duration:** 15 minutes | **Materials:** Printed story cards (cut up)

**Steps:**

1. Give groups 8-10 cards with steps of a story (e.g., “How to make Ragi Mudde”)
2. Cards are shuffled. Groups must arrange them in correct order
3. Some cards may be unnecessary. Identify which ones to remove
4. Discuss: How is this like editing a video?

**CT Skill:** Sequencing, abstraction

## **Cross-Curricular Integration Map**

### **Mathematics Connections**

- **Ch 1:** IP addresses use numbers (decimal/binary conversion)
- **Ch 2:** Pixel coordinates are geometry (x, y positions)

- **Ch 3:** Audio sampling rates involve measurement
- **Ch 4:** Data representation uses statistics (mean, median, mode)
- **Ch 5:** Loops use counters and variables (algebra)
- **Ch 6:** File sizes use units (bytes, kilobytes)

### **Science Connections**

- **Ch 1:** Network signals travel as electromagnetic waves (Physics)
- **Ch 2:** Light and color mixing (RGB model - Physics/Chemistry)
- **Ch 3:** Sound waves and frequency (Physics)
- **Ch 4:** Data collection follows scientific method
- **Ch 5:** Algorithms model real-world processes (Biology, Chemistry)
- **Ch 6:** Software runs on hardware (Physics of electronics)

### **Language Connections**

- **Ch 1:** Writing clear instructions = algorithm design
- **Ch 2:** Visual storytelling through graphics
- **Ch 3:** Audio narration and script writing
- **Ch 4:** Data description and report writing
- **Ch 5:** Program comments = technical writing
- **Ch 6:** Documentation and user guides

## **Digital Citizenship Module**

### **Class 6: Being Safe Online**

**Duration:** 2-3 lessons (can be integrated into Ch 1)

#### **Lesson: Passwords and Privacy**

- Create strong passwords using a formula: Favorite food + birth year + special character
- Practice: “Ragi2014!” vs “password123”
- Rule: Never share passwords with anyone except parents/teachers

#### **Lesson: Identifying Fake News**

- Show examples of fake WhatsApp forwards (use local language examples)
- Check: Is this from a known source? Does it seem too shocking?
- Rule: “When in doubt, don’t share”

#### **Lesson: Screen Time Balance**

- Suggest: 30 minutes screen time, then 10 minutes looking at green trees
- Create a “Screen Time Chart” in their notebook
- Rule: Eyes need rest, body needs movement

### **Class 7: Being Responsible Online**

**Duration:** 2-3 lessons (can be integrated into Ch 6)

#### **Lesson: Cyberbullying Prevention**

- What to do if someone sends mean messages: Tell a trusted adult
- What NOT to do: Don’t reply, don’t forward, don’t keep it secret
- Create a “Help Card” with contact numbers

#### **Lesson: Digital Footprint**

- Everything you post online stays forever
- Exercise: Google yourself (or your school). What do you find?
- Rule: “Don’t post anything you wouldn’t want your teacher to see”

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

# ICT Level 2

## Teaching Guide

Level: Level 2

Class: Class 7

Duration: June - January (8 months)

Period: 40 minutes

ICT Teaching Resources Portal

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## Curriculum Overview

ICT Level 2 builds on the foundations established in Level 1, designed for Class 7 students in KREIS residential schools. The curriculum follows the 5E instructional model and integrates Singapore's Concrete-Pictorial-Abstract approach for deeper conceptual understanding.

## Themes Covered

| Chapter | Theme                    | Periods | Key Topics                           |
|---------|--------------------------|---------|--------------------------------------|
| 1       | Networking and Internet  | 12      | Network types, LAN/WAN,              |
| 2       | Advanced Graphics        | 12      | Internet services, Email,            |
| 3       | Multimedia               | 12      | Image editing, Layers, Filters, GIMP |
| 4       | Data Handling            | 12      | Audio/Video editing, Presentation    |
| 5       | Programming with Scratch | 12      | tools                                |
| 6       | Digital Citizenship      | 12      | Spreadsheets, Charts, Formulas       |
|         |                          |         | Variables, Conditions, Loops         |
|         |                          |         | Online safety, Cyberbullying,        |
|         |                          |         | Digital footprint                    |

## Pedagogical Approach

### 5E Model Implementation

| Phase     | Duration  | Activities                                          |
|-----------|-----------|-----------------------------------------------------|
| Engage    | 5-7 min   | Real-world connection, regional metaphors,          |
| Explore   | 10-12 min | questions                                           |
| Explain   | 10-12 min | Hands-on activity, Driver-Navigator pairs, Lab work |
| Elaborate | 8-10 min  | Live coding, Concept introduction, C-P-A approach   |
| Evaluate  | 5-8 min   | Extended activity, Parson's problems, Application   |
|           |           | Formative check, Exit tickets, Self-assessment      |

### International Pedagogies Used

- **Singapore C-P-A:** Concrete manipulatives to Visual blocks to Abstract code
- **UK NCCE:** Unplugged Computing, Live Coding, Parson's Problems
- **Driver-Navigator:** Pair programming for lab sessions
- **Rural Scaffolding:** Regional metaphors (grain sorting, bus routes, hostel inventories)

### Lesson Plan Structure (40 Minutes)

| Time      | Phase     | Teacher Activity                                    | Student Activity                                 |
|-----------|-----------|-----------------------------------------------------|--------------------------------------------------|
| 0-5 min   | Engage    | Present real-world scenario using regional metaphor | Discuss, share observations                      |
| 5-15 min  | Explore   | Guide hands-on exploration in pairs                 | Work in Driver-Navigator pairs, explore software |
| 15-25 min | Explain   | Live coding demonstration, introduce concepts       | Follow along, ask questions, take notes          |
| 25-35 min | Elaborate | Assign extension activity or Parson's problem       | Apply learning, solve problems independently     |
| 35-40 min | Evaluate  | Conduct formative assessment, assign homework       | Complete exit ticket, receive dormitory task     |

## Assessment Framework

### Assessment Components (Per Class)

| Assessment Type          | Count     | Marks Each | Total      |
|--------------------------|-----------|------------|------------|
| Formative (FA-1 to FA-4) | 4         | 30         | 120        |
| Unit Tests               | 4         | 10         | 40         |
| Summative (SA-1, SA-2)   | 2         | 40         | 80         |
| Practical Lab Exams      | 2         | 20         | 40         |
| <b>Grand Total</b>       | <b>12</b> |            | <b>280</b> |

## Resources Required

### Hardware

- Computer lab with 1 PC per 2 students (minimum)
- Internet connection (2 Mbps minimum)

- Projector for demonstrations
- Speakers and headphones

### Software (Free/Open Source)

- Operating System: Ubuntu Linux or Windows
- Office Suite: LibreOffice / Google Workspace
- Graphics: GIMP, Inkscape
- Audio: Audacity
- Video: Kdenlive
- Programming: Scratch
- Browser: Firefox / Chromium

### Dormitory Tasks (Zero Lab Access)

For residential schools with limited lab time, these paper-and-pencil tasks can be assigned for supervised evening study:

- **Code Dry-Running:** Trace Scratch programs on paper
- **Variable State Tables:** Track variable values through program execution
- **Flowchart Drawing:** Create flowcharts for algorithms
- **Network Diagrams:** Draw and label network topologies on paper
- **Spreadsheet Planning:** Design spreadsheet layouts with formulas on paper
- **Binary Conversion:** Practice decimal to binary conversions
- **Cyberbullying Scenarios:** Discuss and write responses to online safety scenarios
- **Digital Footprint Journal:** Reflect on personal online activities

### Capstone Projects

#### Project 1: School Network Design

**Integration:** Science + ICT

**Description:** Students design a network layout for a hypothetical school campus, identifying devices, connections, and resource sharing requirements. They create network diagrams and present their proposals.

**Skills:** Network topologies, LAN/WAN concepts, diagramming, presentation

#### Project 2: Community Data Dashboard

**Integration:** Mathematics + ICT

**Description:** Students collect data about their community (population, rainfall, agricultural produce) and create a spreadsheet with charts and formulas to visualize and analyze the information.

**Skills:** Spreadsheet formulas, chart creation, data analysis, digital presentation

### Unplugged Activities Library

#### Activity 1: Network Relay (Ch 1 - Networking)

**Duration:** 20 minutes | **Materials:** Paper, tape

**Steps:**

1. Students form a “network” - some are computers, some are routers, some are servers
2. Pass messages (paper slips) from one “computer” to another through “routers”
3. If a “router” is busy (student sits down), messages must find another path
4. Discuss: What happens when parts of a network fail?

**CT Skill:** Network thinking, redundancy, decomposition

#### Activity 2: Layer Editing Puzzle (Ch 2 - Graphics)

**Duration:** 15 minutes | **Materials:** Tracing paper, colored pencils

**Steps:**

1. Draw a simple landscape on one sheet (background)
2. On tracing paper, draw only the tree (layer 1)
3. On another tracing paper, draw only the house (layer 2)
4. Stack them. Discuss: Why do graphic artists use layers?

**CT Skill:** Abstraction, decomposition

**Activity 3: Audio Editing Cards (Ch 3 - Multimedia)**

**Duration:** 15 minutes | **Materials:** Printed story cards

**Steps:**

1. Give groups a story told in 10 cards
2. Task: Remove 2 cards that are not essential to the story
3. Task: Reorder 3 cards to make the story more interesting
4. Discuss: How is this like editing audio/video?

**CT Skill:** Abstraction, pattern recognition

**Activity 4: Algorithm Recipe (Ch 5 - Programming)**

**Duration:** 15 minutes | **Materials:** Paper, pencil

**Steps:**

1. Write a recipe for making chai - but use only these words: START, STOP, REPEAT, IF, THEN
2. The recipe must include: boiling water, adding tea leaves, adding milk, checking taste
3. Trade with another group and follow their recipe exactly
4. Discuss: What happens if steps are missing or unclear?

**CT Skill:** Algorithm design, debugging

**Activity 5: Digital Footprint Audit (Ch 6 - Digital Citizenship)**

**Duration:** 20 minutes | **Materials:** Paper, pencil

**Steps:**

1. List every app, website, and social media you use
2. For each, write: What information does it have about you?
3. Draw a “footprint” showing all your digital traces
4. Discuss: Who can see all this information?

**CT Skill:** Abstraction, evaluation

**Cross-Curricular Integration Map****Mathematics Connections**

- **Ch 1:** Network protocols use binary numbers and addressing
- **Ch 2:** Color theory involves ratios and percentages
- **Ch 3:** Audio/video editing uses timelines (number line concept)
- **Ch 4:** Data handling uses statistics (mean, median, mode, range)
- **Ch 5:** Programming variables are algebraic expressions
- **Ch 6:** Digital ethics involves logical reasoning (IF-THEN)

**Science Connections**

- **Ch 1:** Electromagnetic waves carry network signals (Physics)
- **Ch 2:** Light and color perception (Biology/Physics)
- **Ch 3:** Sound waves and human hearing (Physics/Biology)
- **Ch 4:** Data collection follows scientific method
- **Ch 5:** Algorithms model chemical reactions and biological processes
- **Ch 6:** E-waste and environmental science

## **Language Connections**

- **Ch 1:** Technical writing: network documentation
- **Ch 2:** Visual literacy: interpreting infographics
- **Ch 3:** Script writing for presentations
- **Ch 4:** Data description and report writing
- **Ch 5:** Code comments and documentation
- **Ch 6:** Persuasive writing on digital ethics topics

## **Digital Citizenship Module**

### **Class 7: Being Responsible Online**

**Duration:** 3-4 lessons (integrated throughout the year)

#### **Lesson: Password Security**

- Create passphrases: “MyHostelHas20Rooms!” is better than “password123”
- Practice creating strong passwords for different accounts
- Rule: Different accounts need different passwords

#### **Lesson: Identifying Misinformation**

- Analyze 3 WhatsApp forwards: which are real, which are fake?
- Check: Source, date, multiple sources confirming, reverse image search
- Rule: “Verify before you share”

#### **Lesson: Cyberbullying Response Plan**

- Create a 3-step response: Ignore → Block → Report to adult
- Practice role-playing scenarios
- Create a class “Safety Pledge” poster

#### **Lesson: Digital Footprint Management**

- Google yourself or your school. What information is public?
- Think before you post: Would you show this to your grandparents?
- Rule: “Your digital footprint is your permanent record”

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

# ICT Level 3

## Teaching Guide

Level: Level 3

Class: Classes 8, 9, 10

Duration: June - January (8 months)

Period: 40 minutes

ICT Teaching Resources Portal

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## Curriculum Overview

ICT Level 3 covers advanced topics for Classes 8 through 10. The same textbook is used across all three classes, with differentiated instruction tailored to the developmental stage and academic expectations of each class.

### Themes Covered

| Chapter                                     | Periods | Key Topics                                         |
|---------------------------------------------|---------|----------------------------------------------------|
| Ch 1: Advanced Programming Concepts         | 12      | Functions, Arrays, String operations, Nested loops |
| Ch 2: Data Structures                       | 12      | Lists, Dictionaries, Tuples                        |
| Ch 3: Database Fundamentals                 | 12      | Database concepts, SQL basics                      |
| Ch 4: Web Technologies                      | 12      | HTML basics, CSS introduction                      |
| Ch 5: Algorithms and Computational Thinking | 12      | Algorithm design, Pseudocode                       |
| Ch 6: Emerging Technologies                 | 12      | AI basics, IoT, Cloud computing                    |

## Differentiated Pedagogy by Class

### Class 8 (Foundation)

Class 8 students require heavy Concrete-Phase-Abstract (C-P-A) scaffolding and guided practice. Instruction should rely on visual aids, step-by-step demonstrations, and structured worksheets. Peer collaboration is encouraged through pair programming and group tasks. Concepts are introduced with real-world analogies before transitioning to abstract representations.

### Class 9 (Application)

Class 9 students benefit from a balanced approach between guided and independent learning. They are expected to apply previously learned concepts to new contexts with decreasing levels of teacher scaffolding. Emphasis is placed on debugging skills, logical reasoning, and collaborative problem-solving. Students begin to work on mini-projects with defined parameters.

### Class 10 (Mastery)

Class 10 students focus on independent problem-solving and board exam preparation. Instruction emphasizes revision, application of concepts under timed conditions, and analysis of past examination papers. Students are expected to work with minimal scaffolding and demonstrate mastery through capstone projects and self-directed inquiry.

## 5E Model Implementation

| Phase     | Implementation Strategy                                                                                                                                                     |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Engage    | Activate prior knowledge through demonstrations, thought-provoking questions, or short videos related to the topic. Present a real-world problem that motivates the lesson. |
| Explore   | Students engage in hands-on activities such as coding exercises, database queries, or web page construction. Teacher facilitates but does not direct.                       |
| Explain   | Teacher-led discussion clarifies key concepts, introduces terminology, and formalises rules observed during exploration. Students articulate their understanding.           |
| Elaborate | Students apply learning to new scenarios, extend existing programs, or tackle extension tasks. Emphasis on transfer of knowledge.                                           |
| Evaluate  | Formative assessment through observation, questioning, peer review, and exit tickets. Summative assessment through chapter-end exercises and projects.                      |

## Lesson Plan Structure (40 Minutes)

| Time      | Phase     | Activity                                                                      |
|-----------|-----------|-------------------------------------------------------------------------------|
| 0-5 min   | Engage    | Hook activity: demonstration, question, or quick recall quiz                  |
| 5-15 min  | Explore   | Hands-on coding or activity worksheet; students work individually or in pairs |
| 15-25 min | Explain   | Teacher clarifies concepts, introduces terminology, addresses misconceptions  |
| 25-35 min | Elaborate | Extension task or application problem; independent or group work              |
| 35-40 min | Evaluate  | Exit ticket, quick quiz, or self-reflection prompt                            |

## Assessment Framework

| Assessment Type | Weightage | Components                                                                           |
|-----------------|-----------|--------------------------------------------------------------------------------------|
| Formative       | 40 marks  | Class participation, Worksheets, Homework, Quizzes, Projects                         |
| Summative       | 60 marks  | Mid-term exam, End-of-term exam, Practical assessment                                |
| Total           | 280 marks | Continuous assessment across three terms; final grade derived from cumulative scores |

## Resources Required

- Computers with Python installed (minimum one per pair of students)
- Textbook: ICT Level 3 (as prescribed by the state board)
- Printed worksheets and activity sheets for each chapter
- Projector and screen for demonstrations
- Access to online coding environments as backup (e.g., Replit, Trinket)
- Chart paper, markers, and sticky notes for C-P-A activities
- Database software (SQLite or equivalent for Chapter 3)
- Web browsers with developer tools enabled (for Chapter 4)

## Dormitory Tasks

Dormitory tasks are assigned for evening self-study in the boarding facility. These tasks are designed to reinforce the day's learning and promote independent practice.

- **Code tracing exercises:** Students trace through given code snippets and predict output without running the program.
- **Debugging worksheets:** Intentionally flawed programs are provided for students to identify and correct errors.
- **Concept mapping:** Students create visual maps connecting related concepts from the chapter.
- **Reading assignments:** Selected sections of the textbook are assigned for pre-reading before the next lesson.
- **Mini-projects:** Short guided projects (e.g., building a calculator, creating a simple web page) that consolidate multiple concepts.

## Capstone Projects

Capstone projects are assigned at the end of each term. They require students to integrate knowledge from multiple chapters and demonstrate cumulative understanding.

## Project Guidelines

- Projects must address a real-world problem or scenario relevant to the student's context.
- Students may work individually or in groups of up to three.



- Each project requires a written component documenting the design process, algorithm, and reflections.
- Projects are presented to the class in the final week of the term.

## Sample Project Topics

- **Library Management System:** A console-based program using functions, lists, and basic database operations to manage book records.
- **Personal Portfolio Website:** A multi-page website using HTML and CSS showcasing student work and interests.
- **Traffic Signal Simulator:** A Python program using loops and conditional logic to simulate traffic light sequences.
- **Data Analysis Report:** Students collect survey data, store it in a database, and generate a summary report using SQL queries.

## Unplugged Activities Library

### Activity 1: Sorting Algorithm Race (Ch 2 - Data Structures)

**Duration:** 20 minutes | **Materials:** Numbered cards (1-20)

#### Steps:

1. Give each group 20 shuffled number cards
2. Task: Sort them in ascending order using ONLY these rules:
  - You can only look at 2 cards at a time
  - You can swap 2 cards (but count each swap)
3. Try: Bubble sort vs. Selection sort. Which is faster?
4. Discuss: Why does algorithm efficiency matter?

**CT Skill:** Algorithm design, pattern recognition, evaluation

### Activity 2: Database on Paper (Ch 3 - Databases)

**Duration:** 20 minutes | **Materials:** Graph paper, pencil

#### Steps:

1. Create a “Student” table with columns: Roll No, Name, Class, Marks
2. Fill in 10 rows of data
3. Answer queries: “Who scored above 80?”, “How many students are in Class 8?”
4. Discuss: How is this different from scrolling through a list?

**CT Skill:** Abstraction, data organization

### Activity 3: HTML on Paper (Ch 4 - Web Technologies)

**Duration:** 15 minutes | **Materials:** Paper, pencil

#### Steps:

1. Write the HTML for a simple page: heading, paragraph, image, link
2. Draw what the page would look like in a browser
3. Trade with another group and “render” their HTML mentally
4. Discuss: What does the browser actually do?

**CT Skill:** Abstraction, notation systems

### Activity 4: Flowchart Relay (Ch 5 - Algorithms)

**Duration:** 20 minutes | **Materials:** Flowchart symbol cards

**Steps:**

1. Give each group a set of flowchart symbols (start, process, decision, end)
2. Task: Create a flowchart for “What to eat for lunch?”
3. Include at least 2 decision points
4. Trade and follow another group’s flowchart exactly
5. Discuss: How do computers make decisions?

**CT Skill:** Algorithm design, logical reasoning

**Activity 5: Ethical Dilemma Discussion (Ch 6 - Emerging Tech)**

**Duration:** 20 minutes | **Materials:** Scenario cards

**Steps:**

1. Present scenarios: “Should a self-driving car save the passenger or a pedestrian?”
2. “Should AI be allowed to grade your exams?”
3. Students debate in groups and present their reasoning
4. Discuss: Who is responsible when technology makes mistakes?

**CT Skill:** Evaluation, ethical reasoning

**Cross-Curricular Integration Map****Mathematics Connections**

- **Ch 1:** Time complexity uses mathematical notation (Big-O)
- **Ch 2:** Arrays use indexing (coordinate geometry)
- **Ch 3:** Database normalization uses set theory
- **Ch 4:** CSS uses coordinate systems (box model)
- **Ch 5:** Algorithm analysis uses logarithms and exponents
- **Ch 6:** AI uses probability and statistics

**Science Connections**

- **Ch 1:** Exception handling models error correction in biological systems
- **Ch 2:** Data structures mirror natural organizations (tree structures in biology)
- **Ch 3:** Database relationships model scientific classification
- **Ch 4:** Web protocols use electromagnetic signals (Physics)
- **Ch 5:** Algorithms model natural processes (photosynthesis, circulation)
- **Ch 6:** Emerging tech intersects with all sciences

**Language Connections**

- **Ch 1:** Technical documentation and code commenting
- **Ch 2:** Data description and analysis writing
- **Ch 3:** SQL queries are structured language
- **Ch 4:** Web content requires clear, concise writing
- **Ch 5:** Algorithm descriptions require precise language
- **Ch 6:** Ethics essays and persuasive writing

**Progressive Programming Pathway****Block-Based to Text-Based Transition**

The Level 3 curriculum bridges visual programming (Scratch) to text-based programming (Python).

### **Class 8: Foundation**

- **Term 1:** Scratch advanced features (clones, custom blocks, lists)
- **Term 2:** Python Turtle Graphics (visual transition)
- **Term 3:** Python basics (variables, loops, conditionals)

### **Class 9: Application**

- **Term 1:** Python functions and file handling
- **Term 2:** Data structures in Python (lists, dictionaries)
- **Term 3:** Database operations with Python

### **Class 10: Mastery**

- **Term 1:** Full Python projects + web development basics
- **Term 2:** Integration projects combining all skills
- **Term 3:** Board exam preparation with previous year papers

### **Transition Strategy**

1. Start with Scratch blocks that map directly to Python:
  - repeat 10 → for i in range(10):
  - if on edge bounce → while True:
  - set x to 0 → x = 0
2. Use Python Turtle for visual output (immediate feedback)
3. Gradually introduce text-based debugging
4. Pair Scratch projects with equivalent Python projects

## **Digital Citizenship Module**

### **Class 8-10: Advanced Digital Citizenship**

#### **Lesson: AI Ethics and Bias**

- What is algorithmic bias? Show examples (facial recognition, hiring algorithms)
- Discuss: Can a computer be racist? Sexist?
- Rule: “AI reflects the data it’s trained on - and data reflects human bias”

#### **Lesson: Data Privacy**

- Read the terms of service of a popular app (simplified version)
- What data does it collect? Who can see it?
- Create a “Privacy Checklist” for evaluating apps

#### **Lesson: Online Reputation**

- Google yourself or your school. What do potential employers/colleges see?
- Build a positive digital presence: portfolio, blog, GitHub
- Rule: “Your online presence is your extended resume”

#### **Lesson: Cyber Law in India**

- Overview of IT Act 2000 and its key provisions
- What is illegal online? (hacking, cyberbullying, identity theft)
- Know your rights and responsibilities as a digital citizen